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Unit – 2 : Session – 5 : SLO – 1 | Memory Locations and Addresses

■ Section A – Match the Following (Answers)

1. Word Length → D. Number of bits grouped and accessed together
2. Byte → C. A group of 8 bits
3. Main Memory → A. Collection of storage locations with unique identifiers
4. Address → E. Unique identifier for a word in memory
5. Address Space → B. Minimum number of bits needed to represent all memory locations

■ Section B – Fill in the Blanks

- The term BYTE refers to a group of 8 bits.
- A word of 16 bits is commonly referred to as a HALF word.
- The total number of uniquely identifiable locations in memory is called the ADDRESS SPACE.
- Each storage location in memory has a unique identifier called an ADDRESS.
- To access 32 MB of memory byte-wise, we need 25 bits for addressing ($32 \text{ MB} = 2^{25} \text{ bytes}$).

■ Section C – Activity Task

- Task 1a: 64 MB memory, word = 4 bytes → number of words = $64\text{M}/4 = 16\text{M} = 2^{24}$ → 24 bits are needed to address a word.
- Task 1b: 1 GB memory, word = 8 bytes → number of words = $2^{30} / 2^3 = 2^{27}$ → 27 bits are required to address each word.
- Task 2 – Memory layout (described): a column of words from address 0 (first word) to $2^k - 1$ (last word); each row is one n-bit word; the i-th word sits at address i; addresses label the rows, the n-bit contents are the data.

■ Section D – True or False

- A byte is made up of 16 bits. — FALSE (8 bits)
- Memory addresses are expressed using unsigned binary numbers. — TRUE
- The main memory stores data in single bits only. — FALSE
- The term 'address space' refers to the number of address lines available. — FALSE
- In computer hardware, names are used to identify memory locations. — FALSE (binary addresses are used)